



FLIGHT BIOMIMICRY

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Seaside Sustainability



Inspiration of Flight Biomimicry



Strategies

V-shaped formation that is seen during migrating birds for fuel-efficient flight patterns for long haul flights.

Wing to Plane Wings

Elongated wings of birds like eagles and albatrosses were the inspiration behind the design for gliders and long-range airplanes

- because of lift and endurance over distance with little effort

Short and pointed wings of birds like falcons and hawks were the inspiration behind fighter jets

- because of quick maneuvers and high speed flight

Bird to Balloon

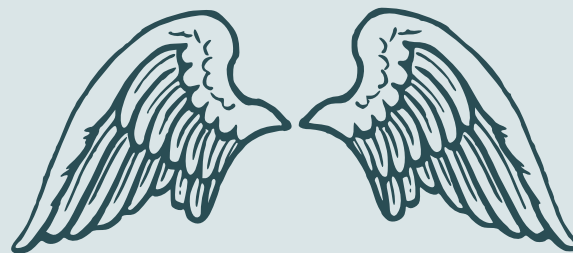
Buoyancy in birds like Humboldt penguins which were the inspiration behind blimps and airships

Rotating Wings

Hovering and flying in small spaces like hummingbirds were the inspiration behind helicopters and drones

Biomimicry

Biomimicry is when people learn from and mimic the strategies living organisms used to replicate during human development of projects.



Inventors

Gliders

Sir George Cayley pretty much created the science behind flight. He was the first to find out the problems of flight that is heavier than air, aerodynamic research, and discovered that curved surfaces lift more than flat ones. He also discovered the streamlining and how to balance aircrafts in flight. Cayley built and flew a model glider and a man carrying glider after.

Airplanes

Idea of flight by Leonardo da Vinci. First airplane invented by the Wright Brothers, which was inspired by pigeons. The first airplane successfully flew in 1903.



Sources:

- [Invention of Flight](#)
- [What is Biomimicry](#)
- [Aviation Designs Inspired by Birds](#)
- [Glider Invention](#)

